Generate | Code | Markdown | Run All | Clear All Outputs | Outline | ... | Select Kernel

```python
#[EXERCISE 3]
customers = [f"{LAST_NAME}_C{i}" for i in range(1, 6)]
product_prices = {
    "NOTEBOOK": 35.00,
    "PEN": 12.00,
    "USB": 350.00,
    "HEADPHONES": 550.00,
    "CALCULATOR": 275.00
}
products = list(product_prices.keys())

transaction_records = []

for i in range(10):
    customer = customers[i % len(customers)]
    product_idx = (i + SEED_NUM + artist_positions[i % len(artist_positions)]) % len(products)
    product = products[product_idx]
    quantity = 1 + ((SEED_NUM + artist_positions[(i + 1) % len(artist_positions)] + i) % 5)
    transaction_records.append({
        "customer": customer,
        "product": product,
        "quantity": quantity,
        "unit_price": product_prices[product],
        "category": "SUPPLIES" if product in ["NOTEBOOK", "PEN"] else "ELECTRONICS"
    })

transaction_records.append(transaction_records[0].copy())

transactions_df = pd.DataFrame(transaction_records)
```

12]

Python

Generate | Code | Markdown | Run All | Clear All Outputs | Outline | ... | Select Kernel

```
print(f"\nGrand total: {transactions_df['transaction_value'].sum():.2f}")
```

[12] Python

```
GENERATED TRANSACTION RECORDS
 customer    product  quantity  unit_price    category  transaction_value
Ganade_C1        PEN         3        12.0    SUPPLIES               36.0
Ganade_C2 HEADPHONES         1       550.0 ELECTRONICS              550.0
Ganade_C3        PEN         1        12.0    SUPPLIES               12.0
Ganade_C4        PEN         3        12.0    SUPPLIES               36.0
Ganade_C5 HEADPHONES         5       550.0 ELECTRONICS             2750.0
Ganade_C1   NOTEBOOK         5        35.0    SUPPLIES              175.0
Ganade_C2   NOTEBOOK         4        35.0    SUPPLIES              140.0
Ganade_C3 CALCULATOR         2       275.0 ELECTRONICS              550.0
Ganade_C4        USB         4       350.0 ELECTRONICS             1400.0
Ganade_C5 CALCULATOR         1       275.0 ELECTRONICS              275.0
Ganade_C1        PEN         3        12.0    SUPPLIES               36.0

ORGANIZED TRANSACTION DATA
 customer    product  quantity  unit_price    category  transaction_value
Ganade_C1   NOTEBOOK         5        35.0    SUPPLIES              175.0
Ganade_C1        PEN         3        12.0    SUPPLIES               36.0
Ganade_C1        PEN         3        12.0    SUPPLIES               36.0
Ganade_C2 HEADPHONES         1       550.0 ELECTRONICS              550.0
Ganade_C2   NOTEBOOK         4        35.0    SUPPLIES              140.0
Ganade_C3 CALCULATOR         2       275.0 ELECTRONICS              550.0
Ganade_C3        PEN         1        12.0    SUPPLIES               12.0
Ganade_C4        PEN         3        12.0    SUPPLIES               36.0
Ganade_C4        USB         4       350.0 ELECTRONICS             1400.0
...
       PEN                4             10      120.00
```